

AP Calculus 2026 Free Response (AB and BC)

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AB/BC 1

Male birds of a certain species arrive at a nesting area over a thirty-day period. The rate at which the male birds arrive at the nesting area at time t days is modeled by a differentiable function M , where $M(t)$ is measured in number of birds per day. Selected values of $M(t)$ are shown in the table.

t (days)	0	5	10	15	20	25	30
$M(t)$ (birds per day)	2	7	16	6	5	2	0

(Note: Your calculator should be in radian mode.)

Part A

Approximate $M'(7.5)$ using the average rate of change of M over the interval $5 \leq t \leq 10$. Show the work that leads to your answer, and indicate units of measure.

$$M'(7.5) \approx \frac{16-7}{10-5} = \frac{9 \text{ birds}}{5 \text{ day}^2}$$

Part B

- (i) Use a midpoint Riemann sum with the three subintervals $[0, 10]$, $[10, 20]$, and $[20, 30]$ to approximate $\int_0^{30} M(t) dt$. Show the work that leads to your answer.

$$\int_0^{30} M(t) dt \approx 10 \cdot 7 + 10 \cdot 6 + 10 \cdot 2 = 150$$

- (ii) Interpret the meaning of $\int_0^{30} M(t) dt$ in the context of the problem.

The integral represents the number of male birds that arrive at the nesting area over the 30 day period.

Part C

The rate at which female birds of the same species arrive at the same nesting area, in birds per day, is modeled by the function F defined as follows:

$$F(t) = \begin{cases} 0 & \text{for } 0 \leq t < 15 \\ 18 + 16 \sin\left(\frac{\pi}{20}(t + 15)\right) & \text{for } 15 \leq t \leq 45 \end{cases}$$

How many female birds of this species arrive at the nesting area from $t = 15$ to $t = 45$? Show the setup for your calculations, and round your answer to the nearest integer.

$$\int_{15}^{45} F(t) dt = 642$$

Part D

On the interval $15 < t < 30$, the difference in the rates at which male and female birds of this species arrive at the nesting area can be modeled by the differentiable function $D(t) = M(t) - F(t)$, where F is the function defined in part C. Is there a time t in the interval $15 < t < 20$ when $D(t) = 0$? Justify your answer.

D is a continuous function.

$$D(15) = 4 > 0$$

$$D(20) = -1.686 < 0$$

$$D(20) < 0 < D(15)$$

Therefore, by the IVT, $D(c) = 0$ for some c , $15 < c < 20$ so yes, there must be a time when $D(t) = 0$.

AB 2

2. The function f is defined by $f(x) = 1.43^x + 0.57$, and the function g is defined by $g(x) = \frac{14x+12}{x+12}$.

The graphs of f and g intersect at the points $(1, 2)$ and (a, b) , as shown in Figure 1.

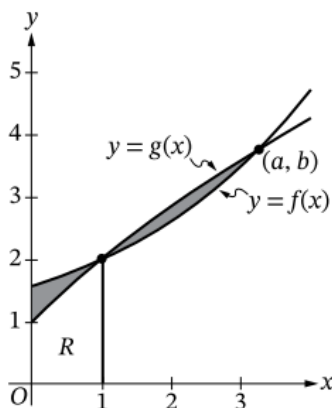


Figure 1

For $x \geq 0$, the equation $y = g(x)$ can be rewritten as $x = h(y) = \frac{12y-12}{14-y}$, where h is a function of y . The graph of $x = h(y)$ and the horizontal line $y = 3.5$ are shown in Figure 2.

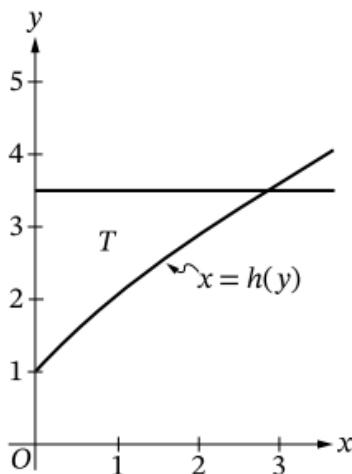


Figure 2

Part A

Let R be the region bounded by the graph of g , the x -axis, the y -axis, and the vertical line $x = 1$, as shown in Figure 1. Find the area of region R . Show the setup for your calculations.

$$f(x) = g(x) \Rightarrow x = 1 \text{ or } 3.256 = a$$

$$b = f(a) = 3.774$$

$$\int_0^1 g(x) dx = 1.513$$

Part B

Region R , described in part A, is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is $\frac{1}{3}$ times the length of its base in region R . Write, but do not evaluate, an integral expression that gives the volume of the solid.

$$\int_0^1 \frac{1}{3} (g(x))^2 dx$$

Part C

The shaded region in Figure 1 is bounded by the graphs of f and g on the interval from $x = 0$ to $x = a$. Find the area of the shaded region. Show the setup for your calculations.

$$\int_0^a |(g(x) - f(x))| dx = 0.631 \text{ or } 0.632$$

Part D

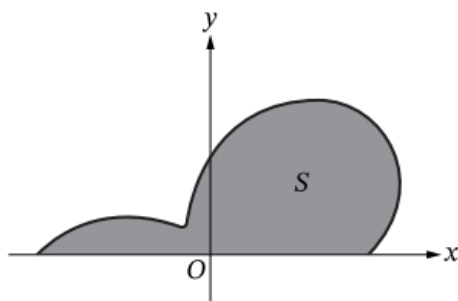
Let T be the region bounded by the graph of $x = h(y)$, the y -axis, and the

horizontal line $y = 3.5$, as shown in Figure 2. Write, but do not evaluate, an integral expression that gives the volume of the solid generated when region T is revolved about the y -axis.

$$\pi \int_1^{3.5} h(y)^2 dy$$

BC 2

Let S be the shaded region bounded by the polar curve $r(\theta) = 3 + 2\sin\theta + 2\cos\theta$, $0 \leq \theta \leq \pi$.



It can be shown that $r'(\theta) = 4\cos\theta - 2\sin\theta$.
(Note: Your calculator should be in radian mode.)

Part A

Find the area of region S . Show the setup of your calculations.

$$\frac{1}{2} \int_0^\pi (3 + 2\sin 2\theta + \cos 2\theta)^2 d\theta = 18.064$$

Part B

There is a point on the curve at which the slope of the line tangent to the curve is $-\frac{3}{7}$. At this point, $\frac{dy}{d\theta} = \frac{3\sqrt{2}}{2}$. Find $\frac{dx}{d\theta}$ at this point. Show the work that leads to your answer.

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{\frac{dy}{d\theta}}{\frac{dy}{dx}} \\ \frac{d\theta}{dx} &= \frac{3\sqrt{2}}{2} \cdot \left(-\frac{7}{3}\right) = -\frac{7\sqrt{2}}{2} \end{aligned}$$

Part C

- (i) Find the value of θ in the interval $0 \leq \theta \leq \frac{\pi}{2}$ at which r has a critical point.

$$r'(\theta) = 4\cos 2\theta - 2\sin 2\theta = 0 \Rightarrow \theta = \frac{1}{2} \tan^{-1} 2 = 0.554$$

- (ii) Use a derivative test to determine whether the critical point is the location of a relative minimum, a relative maximum, or neither for r .

$$r''\left(\frac{1}{2} \tan^{-1} 2\right) = -8.944 < 0. \text{ Because } r'\left(\frac{1}{2} \tan^{-1} 2\right) = 0 \text{ and } r''\left(\frac{1}{2} \tan^{-1} 2\right) < 0, \text{ by the second derivative test there is a relative maximum of } r \text{ when } \theta = \frac{1}{2} \tan^{-1} 2.$$

Part D

Find the average value of $r(\theta)$ on $\frac{\pi}{2} \leq \theta \leq \pi$.

$$r_{ave} = \frac{2}{\pi} \int_{\pi/2}^{\pi} r d\theta = 1.726 \text{ or } 1.727$$

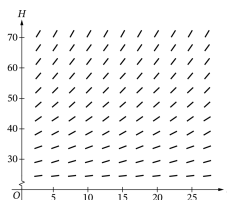
AB/BC 3

A pie is taken from a hot oven. The temperature $H(t)$ satisfies

$$\frac{dH}{dt} = -\frac{1}{15}(H - 20), \quad H(0) = 75.$$

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dH}{dt} = -\frac{1}{15}(H - 20)$.



Because $H > 20$, $\frac{dH}{dt} < 0$, and slope of H would always be negative. However the given slope field displays positive slopes.

Part B

Find the slope of the line tangent to the graph of H at time $t = 0$. Show the work that leads to your answer.

$$\left. \frac{dH}{dt} \right|_{t=0} = -\frac{1}{15}(75 - 20) = -\frac{11}{3}$$

Part C

It can be shown that $\frac{d^2H}{dt^2} = \frac{1}{225}(H - 20)$. The line tangent to the graph of H at time $t = 0$ is used to approximate $H(5)$, the internal temperature of the pie at $t = 5$. Is this approximation an overestimate or an underestimate for the actual value of $H(5)$? Give a reason for your answer.

The tangent line underapproximates $H(5)$ because $\frac{d^2H}{dt^2} > 0$.

Part D Solve for $H(t)$ using separation of variables.

$$\begin{aligned} \int \frac{dH}{H-20} &= -\frac{1}{15} \int dt \\ \ln |H - 20| &= -\frac{t}{15} + C \\ H - 20 &= Ke^{-t/15} \\ H &= 20 + Ke^{-t/15} \\ 75 &= 20 + K \\ H(t) &= 20 + 55e^{-t/15} \end{aligned}$$

AB/BC 4

Let f be twice differentiable on $[-4, 4]$ with $f(2) = 3$. The graph of f' is given.

Part A If $g(x) = f(x) - \ln x$, find $g'(2)$.

$$\begin{aligned} g'(x) &= f'(x) - \frac{1}{x} \\ g'(2) &= 1.5 - \frac{1}{2} = 1 \end{aligned}$$

Part B Find all inflection points of f on $0 < x < 3$.

The graph of f has inflection points at the values of x where f' has relative extrema.

On the open interval $(0, 3)$ these are only $x = 1$.

Part C On what intervals is f increasing and concave down?

The graph of f is increasing and concave down when the graph of f' is positive and decreasing.

This is on the interval $(1, 3)$.

Part D Find where f has an absolute minimum and maximum.

Candidates for the absolute extrema are the endpoints of the interval ($x = -4, 4$) or when f' changes signs ($x = -2$).

At $x = 4$, we have an absolute maximum because $f(4) = 3 + \int_2^4 f'(x)dx > 3$

At $x = -2$, there is an absolute minimum because $f(-2) = 3 - \int_{-2}^2 f'(x)dx < 3$

Note that although $f(-4) > f(-2)$, there is not an absolute maximum at $x = -4$ since visually $\int_{-4}^2 f'(x)dx > 0$ and thus $f(-4) = 3 - \int_{-4}^2 f'(x)dx < 3 < f(4)$.

AB 5

A remote-controlled toy car moves back and forth along a straight path so that its velocity at time t is given by the function v , where $v(t)$ is measured in feet per second and t is measured in seconds.

$$v(t) = \begin{cases} t^4 - 8t^3 + 16t^2 & \text{for } 0 \leq t \leq 4 \\ 0 & \text{for } 4 < t < 6 \\ 10 \cos\left(\frac{\pi}{3}t\right) - 10 & \text{for } 6 \leq t \leq 12 \end{cases}$$

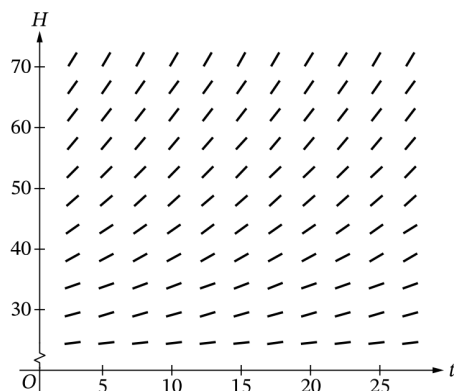
The graph of $v(t)$ is shown.

Part A

Find the acceleration of the car at time $t = 1$ second. Show the work that leads to your answer.

$$a(1) = v'(1) = 4(1)^3 - 24(1)^2 + 32(1) = 12$$

Part B



Is the car speeding up or slowing down at time $t = 1$ second? Give a reason for your answer.

$$v(1) = 1 - 8 + 16 > 0.$$

Because $a(1) > 0$ and $v(1) > 0$, the car is speeding up at $t = 1$.

Part C

Find the distance, in feet, that the car traveled over the time interval $0 \leq t \leq 4$ seconds. Show the work that leads to your answer.

$$\int_0^4 v(t) dt = \left[\frac{1}{5}t^5 - 2t^4 + \frac{16}{3}t^3 \right]_0^4 = \frac{4^5}{5} - 2(4)^4 + \frac{16(4)^3}{3}$$

Part D

Find the average velocity of the car over the time interval $6 \leq t \leq 12$ seconds. Show the work that leads to your answer.

$$\begin{aligned} v_{ave} &= \frac{1}{6} \int_6^{12} v(t) dt \\ &= \frac{1}{6} \left(\frac{30}{\pi} \sin\left(\frac{\pi}{3}t\right) - 10t \right) \Big|_6^{12} \\ &= \frac{1}{6} (-60) \\ &= -10 \end{aligned}$$

BC 5

Let $f(x) = (x - 1)^3$ and $g(x) = e^{2x-4}$. The graphs intersect at $(2, 1)$.

Part A

Evaluate $\int_1^2 f(x) dx$.

$$\int_1^2 f(x) dx = \int_1^2 (x - 1)^{1/3} dx = \frac{3}{4} (x - 1)^{4/3} \Big|_1^2 = \frac{3}{4}$$

Part B

Write an integral for the volume when the region is rotated about the x -axis.

$$\pi \int_1^2 (x-1)^2 dx$$

Part C

Write an integral expression for the perimeter.

$$P = 1 + 1 + \int_1^2 \sqrt{1 + \left(\frac{1}{3}(x-1)^{-2/3}\right)^2} dx$$

Part D

Evaluate $\int_2^3 g(x) dx$.

$$\lim_{b \rightarrow \infty} \int_2^b e^{(-2x+4)} dx = \lim_{b \rightarrow \infty} \left[-\frac{1}{2} e^{(-2x+4)} \Big|_2^b \right] = \frac{1}{2}$$

AB 6

The function f is twice differentiable. The table gives values of f and its derivative f' at selected values of x .

x	0	2	3	6
$f(x)$	-1	3	8	5
$f'(x)$	-5	4	9	-2

Part A Find $\lim_{x \rightarrow 2} \frac{f(x)}{x}$, or state that the limit does not exist.

$$\lim_{x \rightarrow 2} \frac{f(x)}{x} = \frac{f(2)}{2} = \frac{3}{2}$$

Part B

Let $g(x) = f(f(x))$. Find $g'(2)$. Show the work that leads to your answer.

$$\begin{aligned} g'(x) &= f'(f(x)) f'(x) \\ g'(2) &= f'(3) (4) = 8 \cdot 4 = 32 \end{aligned}$$

Part C

Let h be a differentiable function such that $h(0) = 10$ and $h'(x) = f'(3x)$. Find $h(2)$. Show the work that leads to your answer.

$$\begin{aligned} h(2) &= 10 + \int_0^2 f'(3x) dx \\ &= 10 + \frac{1}{3} \int_0^6 f'(u) du \\ &= 10 + \frac{1}{3} (f(6) - f(0)) \\ &= 10 + \frac{6}{3} = 12 \end{aligned}$$

Part D

Let k be the function defined by $k(x) = \int_0^x t^2 f(t) dt$.

(i) Find $k'(x)$.

$$k'(x) = x^2 f(x)$$

(ii) Find $k''(3)$. Show the work that leads to your answer.

$$k''(3) = 2(3) f(3) + 3^2 f'(3) = 6 \cdot 8 + 9 \cdot 9 = 48 + 81 = 129$$

BC 6

The Maclaurin series for a function g is given by

$\sum_{n=0}^{\infty} 2\left(-\frac{x}{5}\right)^n = 2 - \frac{2}{5}x + \frac{2}{25}x^2 - \frac{2}{125}x^3 + \frac{2}{625}x^4 - \dots + 2\left(\frac{-x}{5}\right)^n + \dots$ and converges to $g(x)$ for $|x| < 5$.

Part A

For $x > 0$, the Maclaurin series for g is an alternating geometric series. Find $g(3)$. Show the work that leads to your answer.

$$g(3) = \frac{2}{1+\frac{3}{5}} = \frac{5}{4}$$

Part B

The function f is defined by $f(x) = g'(x)$. Write the first four nonzero terms of the Maclaurin series for f .

$$f(x) = g'(x) \approx -\frac{2}{5} + \frac{4}{25}x - \frac{6}{125}x^2 + \frac{8}{625}x^3$$

Part C

The second-degree Taylor polynomial for f about $x = 0$ is used to approximate $f\left(\frac{5}{2}\right)$ as $-\frac{3}{10}$, where f is the function defined in part B. Justify that $\left|f\left(\frac{5}{2}\right) - \left(-\frac{3}{10}\right)\right| \leq \frac{1}{5}$.

The Maclaurin series for $f(5/2)$ is an alternating series with coefficients that decrease in absolute value to 0. Thus by the Alternating Series Error Bound, $\left|f\left(\frac{5}{2}\right) - \left(-\frac{3}{10}\right)\right| \leq \frac{8}{625}\left(\frac{5}{2}\right)^3 = \frac{1}{5}$

Part D

The function h is defined by $h(x) = 25g(x) - 2e^x$.

(i) Write first three nonzero terms of the Maclaurin series for e^x .

$$e^x \approx 1 + x + \frac{x^2}{2}$$

(ii) Write first three nonzero terms of the Maclaurin series for h .

$$h(x) \approx 2\left(1 + x + \frac{x^2}{2}\right) + 50 - 10x + 2x^2 = 52 - 8x + 3x^2$$